VisionGuard: Crisp Viva Script

From scratch to advanced. Every answer is 1–2 sentences. Actual cane wiring: Trig A0, Echo A1, moisture A2, buzzer D13. S Darshan (25BCE2883) · Tanay Agarwal (25BCE2475) · BACSE324.

Level 1: The 10-second answer

> "VisionGuard is a smart cane for visually impaired people. An ultrasonic sensor beeps faster as an obstacle comes within 50 cm, and a moisture probe at the tip gives a continuous beep on water, which always has priority."

Level 2: The 30-second answer (input → process → output)

> "**Inputs:** the HC-SR04 ultrasonic sensor measures distance, and a resistive probe with an LM393 module senses water. **Process:** an Arduino Nano (ATmega328P) checks water first, then distance. **Output:** an active buzzer gives a continuous tone for water and graded beeps for obstacles. **Power:** one 18650 Li-ion cell through a rocker switch to the 5V pin."

Level 3: Components (one line each)

Component	Crisp answer
Arduino Nano	The controller: ATmega328P, 8-bit, 16 MHz, 32 KB flash, 2 KB SRAM, 10-bit ADC. Small enough for a cane.
HC-SR04	Ultrasonic distance sensor: sends 40 kHz pulses and times the echo. Range 2–400 cm, beam about 15°.
Moisture probe	Two copper tracks. Water conducts, so resistance drops.
LM393 module	Turns the probe's resistance into a voltage on A0 (we use this). D0 is its comparator yes/no output (unused).
Active buzzer	Has its own oscillator, so a HIGH on D13 makes a tone.
18650 cell	Li-ion, 3.7 V nominal (4.2 full, about 3.0 empty), 1200 mAh, 4.44 Wh. IMR = manganese chemistry, safer.
Rocker switch	Breaks the battery's + line: on/off.
220 µF capacitor	Bulk filter across 5V and GND; supplies current spikes when the buzzer switches. Stripe = negative.
Header pins + jumpers	Male pins soldered on the Nano; sensors plug on with female jumpers, so modules can be removed and tested.
PVC pipe	Light, cheap, strong cane body, and an insulator.
Hot glue / cable tie / foam	Hot glue mounts the parts (insulating, slightly flexible); the cable tie holds the battery; foam holds the HC-SR04 straight.
USB cable + CH340	Programming and Serial Monitor; the CH340 chip converts USB to UART.

Level 4: Wiring

From	To Nano	Crisp reason
HC-SR04 Trig (orange)	A0	Output pulse; analog pins also work as digital.
HC-SR04 Echo (yellow)	A1	Input: the echo pulse width.
HC-SR04 VCC/GND (white/black)	ICSP 5V/GND	The 6-pin ICSP header also carries 5V and GND.
LM393 A0 (grey)	A2	Analog moisture value, 0–1023.

LM393 VCC/GND (red/brown)	5V / GND lines	Module power.
Buzzer + / −	D13 / USB shell (GND)	Soldered directly; the on-board "L" LED blinks with it.
Battery + → switch	5V pin	Not VIN: VIN's regulator needs about 7 V.
Battery −	GND	Common ground for every module.
220 µF capacitor	5V and GND	Supply smoothing.
D3	Free	Reserved for a vibration motor via the IRLZ44N MOSFET.

Level 5: Code (every function, one line)

Code part	Crisp answer
setup()	Runs once: sets pin modes, starts Serial at 9600, runs the self-test.
loop()	Runs forever: reads sensors, decides the state, drives the buzzer, prints status.
readDistanceCm()	10 µs pulse on Trig, `pulseIn` on Echo with a 30 ms timeout; distance = time × 0.034 ÷ 2.
pulseIn()	Measures how long a pin stays HIGH, in µs. It busy-waits (blocks).
medianOf3()	Middle of the last 3 distances: removes one wrong echo.
readMoisture()	Average of 4 `analogRead` values: reduces noise.
Hysteresis (500/550)	Wet below 500, dry only above 550: stops flickering.
readSupplyMv()	Measures the internal 1.1 V reference against AVcc: Vcc = 1.1 × 1023 ÷ reading. That's the battery voltage with no extra wires.
enum State	CLEAR, OBSTACLE, WATER: exactly one each loop, water checked first.
map() / constrain()	constrain keeps distance in 2–50 cm; map turns it into a 50–400 ms beep gap.
millis()	Milliseconds since start (Timer0). Used for non-blocking timing instead of `delay()`.
setBuzzer()	Switches the buzzer (and D3) and remembers when, for timing.
selfTest()	At start-up: 5 pings, moisture and battery check, then a ready chime (or one long beep if the sensor fails).
Heartbeat / low battery	When clear: 1 chirp every 30 s (cane is on); 3 chirps every 20 s if below 3.4 V.
Serial.print	Status 5 times a second at 9600 baud, for debugging and calibration.
60 ms ping interval	Datasheet recommendation, so old echoes don't confuse the next reading.

Level 6: Advanced (architecture, physics, power)

Architecture: 8-bit AVR RISC, Harvard (separate flash and SRAM buses), 32 registers, most instructions in 1 cycle (about 16 MIPS), 2-stage pipeline.

Memory: flash holds the program (≈30 KB usable, 2 KB bootloader); SRAM holds variables and the stack; EEPROM (1 KB) could store the threshold.

Code path: avr-gcc → .hex → avrdude → CH340 (USB to UART) → bootloader writes flash. Our Nano has the new bootloader (115200 baud).

ADC: 10-bit successive approximation, 1024 levels, about 4.9 mV/step at 5 V, about 104 µs per conversion.

Ratiometric: the probe divider and the ADC share the supply, so the moisture reading doesn't depend on battery voltage.

Ultrasonic physics: d = v·t ÷ 2, v ≈ 343 m/s = 0.034 cm/µs. 2 cm minimum because of transducer ringing. About 0.2% error per °C.

Moisture physics: $V_{A0} = Vcc \cdot R_p/(R_p+R_f)$. Dry → high R_p → ≈1023; water (dissolved salts) → low R_p → low value. DC causes electrolysis, which is why the probe corrodes.

Software design: non-blocking super-loop, no RTOS, soft real-time. Water reacts within milliseconds because nothing uses `delay()` in the loop.

Power limits: 16 MHz is guaranteed from about 3.8 V, so a drained cell is marginal; the brown-out detector resets at about 2.7 V. A pin gives 20 mA (40 mA max), so a motor needs a MOSFET plus a flyback diode.

Battery life: ≈40 mA → 1200 mAh ÷ 40 mA ≈ 30 h theoretical.

A–Z: every term in the project

ADC	Analog-to-digital converter: voltage → number (0–1023).
Active buzzer	Buzzer with a built-in oscillator; DC makes sound.
AVcc	The ADC's supply pin; our default reference.
Bandgap (1.1 V)	Fixed internal reference, used to measure the battery.
Baud	Serial speed in bits/s: 9600 for the monitor, 115200 for upload.
Bootloader	Small program in flash that accepts code over USB serial.
Brown-out detector	Resets the chip if the voltage drops too low.
Capacitor	Stores charge; smooths supply dips (i = C·dV/dt).
CH340	USB-to-UART chip on clone Nanos.
Cold joint	Dull, weak solder joint from too little heat.
Comparator	Compares two voltages → HIGH/LOW (the LM393).
constrain()	Clamps a value into a range.
delay()	Blocking wait; avoided in our loop.
Echo pin	HC-SR04 output; HIGH for the round-trip time.
EEPROM	1 KB memory that survives power-off.
Electrolysis	DC through water corrodes the probe.
Flash	32 KB program memory.
Flux	Removes oxide so solder flows.
Flyback diode	Absorbs a motor's switch-off voltage spike.
GND	0 V reference, common to all parts.
Harvard architecture	Separate program and data memory/buses.
Header pins	Metal pins soldered on the board for jumpers.
Hysteresis	Two thresholds (on/off) to stop flicker.
ICSP header	6-pin programming header; also gives 5V/GND.
IRLZ44N	Logic-level MOSFET for the future motor.
Jumper wire	Plug-in wire (female end fits header pins).
LM393	Dual comparator on the moisture module.
loop()	Function that repeats forever.
map()	Linear scaling from one range to another.
Median filter	Uses the middle of several readings; rejects outliers.
millis()	Time since start in ms (Timer0 interrupt).
MOSFET	Transistor switch for larger currents.
Non-blocking	Code that never waits, so everything keeps running.
pulseIn()	Measures pulse width in µs; blocking.
PWM	Fast on/off switching to vary power (D3 supports it).
Ratiometric	Reading depends on a ratio, not the supply value.
RISC	Simple instructions, mostly 1 clock each.
Serial / UART	Two-wire serial link (RX/TX) to the laptop.
setup()	Function that runs once at power-on.
Soft real-time	Should respond fast; a small delay isn't catastrophic.

SRAM	2 KB working memory for variables.
Super-loop	Bare-metal design: setup once, loop forever, no OS.
Trig pin	HC-SR04 input; a 10 µs HIGH starts a measurement.
Ultrasound	Sound above 20 kHz (we use 40 kHz).
VCC / 5V pin	Supply rail; our battery feeds it directly.
VIN	Input to the on-board regulator; needs about 7 V+.
Voltage divider	Two resistances; the middle voltage depends on their ratio.

Top 12 questions, crisp answers

Why divide by 2? Sound goes there and back.

Why water first? The tip is already at the danger spot; slipping or falling is the bigger risk.

Why A0/A1 for the ultrasonic? Analog pins also work as digital pins (D14/D15); they were free and close to the jumpers.

Why the 5V pin, not VIN? VIN's regulator needs about 7 V; one cell is 3.7 V.

Why millis(), not delay()? Non-blocking: sensors keep being checked during beeps.

How do you measure the battery? The internal 1.1 V bandgap against AVcc: Vcc = 1.1 × 1023 ÷ reading.

Why 30 ms timeout? 400 cm round trip ≈ 23.5 ms; the default would freeze for 1 s.

Why 60 ms between pings? The datasheet says so, to avoid old echoes.

Why no breadboard? A cane vibrates; soldered power joints and jumpers on header pins are reliable.

Biggest limitation? The narrow beam misses head-level objects, and the probe senses water only on contact.

Next upgrade? A vibration motor on D3 via the IRLZ44N, plus a 5 V boost converter and charger.

Don't know an answer? "I'm not sure, sir. I'd check the datasheet."